Huy Hoang Le

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Education

The University of Danang – University of Science and Technology (DUT). Da Nang, Vietnam Sep 2020 – Aug 2025
Degree of Engineer in Control Engineering and Automation (Specialized In-depth Program, 180 Credits, Master's equivalent).

Professional Experience

Research Engineer, R&D Department, Powermore Ltd. Sep 2025 – Present
Da Nang, Vietnam.
Address: 11/2 Che Lan Vien Street, Ngu Hanh Son District, Da Nang 550000, Vietnam.

Research Interests

Core Domains: Scientific Computing & Operator Learning (DeepONet, PINN) for Power Systems; Prognostics and Health Management (PHM); Digital Twins; Uncertainty Quantification.

Research Collaborations

Purdue University (Remote Research Group) Dec 2025 – Present
Collaborating directly with researchers under the supervision of **Prof. Guang Lin** on operator learning (DeepONet) and physics-informed ML for power system transients and digital twins.

- *Key Outputs:* Lead/co-author of manuscripts under review in *Neural Networks* and *IEEE Transactions on Industrial Informatics*.

The University of Danang – University of Science and Technology 2023 – Present
Leading and collaborating on advanced Prognostics and Health Management (PHM) and reliability engineering tracks for electrical power systems.

- *Key Outputs:* Published work and multiple manuscripts under review in PHM, reliability, and signal-processing venues.

Journal Articles

1. **Huy Hoang Le**, Ba Tu Phung, Dai Huynh, Kim-Anh Nguyen, Transformer Fault Diagnosis Using an Efficient Simulation-Driven Variational Quantum Classifier with Domain-Aware Feature Encoding, *Alexandria Eng. J.* 144 (2026) 58–78. https://doi.org/10.1016/j.aej.2026.04.032.

2. Kim-Anh Nguyen, **Huy Hoang Le**, Ba Tu Phung, An Adaptive Multi-Fuzzy Logic Model for Diagnosing Transformer Faults Using Dynamic Weight Optimization, *e-Prime* 13 (2025) 101048. https://doi.org/10.1016/j.prime.2025.101048.

3. **Huy Hoang Le**, Kim-Anh Nguyen, Generalization Bounds and Sample Complexity for Remaining Useful Life Prediction from Complete Degradation Trajectories, *Meas. Sci. Technol.* 37 (2026) 226203. https://doi.org/10.1088/1361-6501/ae7109.

4. **Huy Hoang Le**, Kim-Anh Nguyen, Charging Phase Health Indicators for Battery State-of-Health Estimation: A Systematic Comparison of CC, CV, and Combined Approaches under Cross-Battery Validation, *Eksploatacja i Niezawodność* 28 (2026) 220211. https://doi.org/10.17531/ein/220211.

5. **Huy Hoang Le**, Kim-Anh Nguyen, Information-Theoretic Limits of Health Indicator

Construction for Remaining Useful Life Prediction, *Eng. Sci. Technol. an Int. J.* 81 (2026) 102485. https://doi.org/10.1016/j.jestch.2026.102485.

6. Kim-Anh Nguyen, **Huy Hoang Le**, Ba Tu Phung, Huy Vu Tran, A Fuzzy Logic Approach Combined with IEC and Roger Methods for Power Transformer Fault Diagnosis Based on DGA, *Danang J. Sci. Technol.* 23 (2025) 64–70. https://doi.org/10.31130/ud-jst.2025.010.

7. Kim-Anh Nguyen, Huynh Dai, Thi Thu Le, **Huy Hoang Le**, Ba Tu Phung, Thermal Analysis of Low-Voltage Power Cables Considering the Presence of Current Harmonics, *Quy Nhon Univ. J. Sci.* 20 (2026) 85–96. https://doi.org/10.52111/qnjs.2026.20307.

8. **Huy Hoang Le**, Haoguang Wang, Christian Moya, Marcos Netto, Guang Lin, Implicit Constrained Learning for Neural Operators on Stiff Differential–Algebraic Systems, *Neural Networks*, (Under review). arXiv:2604.19930.

9. **Huy Hoang Le**, Kim-Anh Nguyen, Fisher Information-Guided Label Design for Remaining Useful Life Prediction: Closing the Gap Between Degradation Physics and Learning Targets, *Mechanical Systems and Signal Processing*, (Under review).

10. **Huy Hoang Le**, Kim-Anh Nguyen, Calibrated Heteroscedastic Regression with Physics-Informed Constraints: A Framework for Uncertainty Quantification in Time-Series Prediction, *Machine Learning: Science and Technology*, (Under review).

11. Dai Huynh, **Huy Hoang Le**, Kim-Anh Nguyen, A Physics-Informed Training-Target Design Framework for Remaining-Useful-Life Prediction, *Arabian Journal for Science and Engineering*, (Under review).

12. **Huy Hoang Le**, Kim-Anh Nguyen, Risk–waste geometry for condition-based replacement: Optimality and finite-fleet estimation, *Reliability Engineering & System Safety*, (Under review).

13. **Huy Hoang Le**, Kim-Anh Nguyen, Information loss of fixed linear health indicators under competing degradation mechanisms and variable operating conditions, *Reliability Engineering & System Safety*, (Under review).

14. **Huy Hoang Le**, Dai Huynh, Kim-Anh Nguyen, Post-hoc Constrained Conformal Prediction for Prognostics: Feasible Intervals that Compose with Adaptive Calibration under Distribution Shift, *Heliyon*, (Under review).

15. Dai Huynh, **Huy Hoang Le**, Kim-Anh Nguyen, A viability-based composite health index for oil-immersed transformers using physics-constrained cascaded operator decomposition, *Engineering Science and Technology, an International Journal*, (Under review).

16. Haoguang Wang, **Huy Hoang Le**, Akhila Kandivalasa, Christian Moya, Marcos Netto, Guang Lin, Unified Mamba–MoE Surrogate for Closed-Loop Simulation and Measurement-Window Forecasting of Inverter Transients, *IEEE Transactions on Industrial Informatics*, (Under review). arXiv:2608.15051.

17. Quang Vu Truong, **Huy Hoang Le**, Dai Huynh, Kim-Anh Nguyen, Normalization protocol dominates posterior expressiveness in calibrated physics-informed Bayesian prediction of IGBT remaining useful life, *Microelectronics Reliability*, (Under review).

18. Dai Huynh, **Huy Hoang Le**, Kim-Anh Nguyen, A Two-Axis Review of Nuclear Reliability and Risk Assessment, with Implications for a Newcomer VVER-1200 Program, *Progress in Nuclear Energy*, (Under review).

19. **Huy Hoang Le**, Kim-Anh Nguyen, Indicator-aware design of run-to-failure degradation tests for data-driven prognostics, *Reliability Engineering & System Safety*, (Under review).

20. **Huy Hoang Le**, Kim-Anh Nguyen, Segmented Projections of a Rotating Signal Subspace: Exact Solution, Ky Fan Geometry, and a Sharp Rate Law, *IEEE Transactions on Signal Processing*, (Under review).

21. **Huy Hoang Le**, Kim-Anh Nguyen, A cumulative Fisher-information framework for the timing of health-indicator information in prognostics, *Mechanical Systems and Signal Processing*, (Under review).

Conference Articles

1. **Huy Hoang Le**, Quang Vu Trương, Kim-Anh Nguyen, Quantum-Assisted Policy Learning for Fast Volt/Var Control in Distribution Networks, *2026 8th Vietnam Conference on Control and Automation (VCCA)*, 2026.

2. Quang Vu Truong, **Huy Hoang Le**, Kim-Anh Nguyen, Reliability Analysis of Gas-Insulated Switchgear Using Hybrid Fault Tree, *Proc. 2026 11th Int. Conf. on Applying New Technology in Green Buildings (ATiGB)* (2026) 1256–1261. https://doi.org/10.1109/ATIGB70203.2026.11628406.

3. Kim-Anh Nguyen, **Huy Hoang Le**, Ba Tu Phung, Optimized Fuzzy Logic Approach with the IEEE Key Gas Method for Diagnosing Power Transformer Faults Using Dissolved Gas Analysis, *Proc. 2025 10th Int. Conf. on Applying New Technology in Green Buildings (ATiGB)* (2025) 114–119. https://doi.org/10.1109/ATiGB66719.2025.11142211.

4. Kim-Anh Nguyen, **Huy Hoang Le**, Ba Tu Phung, Huy Vu Tran, Enhancing Data Preprocessing Layer for Power Transformer Fault Diagnosis Using DGA Combining Fuzzy Logic and Duval Triangle 1, *Proc. 2025 8th Int. Conf. on Circuits, Systems and Simulation (ICCSS)* (2025) 94–98. https://doi.org/10.1109/ICCSS65911.2025.11081751.

Achievements

Third Prize, Faculty-level Scientific Research, 2025.

Bronze Medal, Southern Regional Chemistry Olympiad, 2017.